

CSC 315 Week One TTTT2

Condensed Study Notes

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CSC 315 : COMPUTER ARCHITECTURE AND ORGANIZATION I

This course, CSC 315: Computer Architecture and Organization I, introduces the fundamental principles of how computer systems are designed and how they operate.

It delves into the organization of computer hardware and the architecture that dictates how software interacts with that hardware. Understanding these concepts is key to comprehending the performance and capabilities of computing devices.

Week One: Lesson One

This section marks the beginning of CSC 315, an introductory course to computer architecture and organization. It focuses on the foundational principles of hardware design and software interaction. The course aims to build a strong understanding of how computers work at a fundamental level.

Introduction

This course, CSC 315, delves into computer architecture and organization, focusing on the core principles of how hardware is designed and how it interacts with software.

Key concepts introduced:

- **Computer Architecture and Organization:** The study of a computer's fundamental attributes and operational units, and their implementation in hardware.
- **Von Neumann Architecture:** A foundational model for computer design where program instructions and data share the same memory space and are processed sequentially by a single CPU.
- **Fundamental Building Blocks (Structure and Function):** Understanding the basic components of a computer system and how they operate to perform tasks.

1.1 Organization and Architecture

Computer architecture refers to attributes visible to a programmer that directly impact program execution.

Key architectural attributes include:

- The instruction set.
- Data type representation (e.g., number of bits for characters).
- I/O mechanisms.
- Memory addressing techniques.

Instruction Set Architecture (ISA) is often used interchangeably with computer architecture. The ISA defines instruction formats, opcodes, registers, memory, instruction execution effects, and control

algorithms.

Computer organization refers to the operational units and their interconnections that implement architectural specifications. These are hardware details often transparent to the programmer, such as control signals, peripheral interfaces, and memory technology.

An architectural decision is *whether* a computer has a multiply instruction. An organizational decision is *how* that instruction is implemented (e.g., a dedicated multiply unit vs. repeated use of an add unit).

Manufacturers often offer product families with the same architecture but different organizations (e.g., IBM System/370). For microcomputers, architecture and organization are closely linked, with technological changes influencing both.

1.2 Structure and Function

To understand complex systems like computers, we use a hierarchical approach. This means breaking down a system into interrelated subsystems, and then further breaking down those subsystems until we reach the most basic components.

At each level of this hierarchy, we consider two key aspects:

- **Structure:** How the different components are connected and arranged.
- **Function:** What each individual component does as part of the overall structure.

This top-down approach is generally the clearest way to describe and design complex systems.

Structure (The Internal structure of the Computer)

A traditional single-processor computer has four main structural components:

- **Central Processing Unit (CPU):** Controls the computer's operations and performs data processing. It's often called the processor.
- **Control Unit:** Manages the CPU's operations and, by extension, the entire computer.
- **Arithmetic and Logic Unit (ALU):** Carries out the computer's data processing tasks.
- **Registers:** Provide immediate storage within the CPU.
- **Main Memory:** Stores data.
- **I/O (Input/Output):** Manages data transfer between the computer and its external environment.
- **System Interconnection:** A mechanism, often a system bus, that allows communication between the CPU, main memory, and I/O.

Within the CPU, there's also a **CPU interconnection** which facilitates communication between the control unit, ALU, and registers.

Modern computers may use multiple processors, deviating from the traditional single-processor model. Parallel and pipelined techniques add complexity to computer organization.

A microprogrammed control unit is one approach where functionality is defined by executing microinstructions.

Function

Computers perform four basic functions:

- **Data processing:** Manipulating data through operations like counting, swapping, adding, and comparisons.
- **Control:** A control unit directs the computer's resources and coordinates its parts based on instructions.
- **Data storage:** Temporarily holding data currently being processed (short-term) and permanently storing files for later use (long-term).
- **Data movement:** Transferring data. This includes **input/output (I/O)**, where data moves between the computer and directly connected **peripheral** devices, and **data communications**, where data moves over longer distances to remote devices.

multicore computer structure

Modern computers are often multicore, meaning they have multiple processors on a single chip.

Key terms:

- **Processor:** A silicon component that interprets and executes instructions. Can contain one or more cores.
- **Multicore Processor:** A processor containing multiple cores.
- **Core:** An individual processing unit within a processor chip. A core can function like a CPU in a single-CPU system, or be specialized (e.g., for vector operations).

These multicore processors are typically housed on a motherboard, which is a printed circuit board (PCB) that interconnects chips and components. The motherboard contains slots for the processor chip, memory chips, I/O controllers, and expansion boards.

A typical multicore processor chip includes multiple cores and an L3 cache, which is usually accessible to all cores via control circuits. Additional logic manages operations between cores, the cache, and the motherboard's external circuitry.

Single Core Structure

A single core is a functional unit within a processor chip.

Key components of a core:

- **Instruction Logic:** Fetches and decodes instructions, identifying the operation and operand locations.
- **Arithmetic and Logic Unit (ALU):** Executes the operations specified by instructions.

- **Load/Store Logic:** Manages data transfers between main memory and the core, utilizing cache.
- **L1 Cache:** A small, fast memory within the core, split into an Instruction Cache (I-cache) for instructions and a Data Cache for operands/results.
- **L2 Cache:** Often included as part of the core; can be split (instruction/data) or combined.

These functional elements are general concepts; their physical layout can vary, especially if implemented with microprogrammed control units.

Other components on a processor chip, not necessarily within a single core, include:

- Processor sockets
- PCI-Express slots
- Serial ATA (SATA) sockets
- DDR main memory interfaces
- Ethernet controller and ports
- USB sockets
- Chipsets (e.g., Intel 3420) for I/O control and direct memory access.

1.3 Von Neumann architecture

The Von Neumann architecture, described by John Von Neumann in the 1940s, outlines the fundamental relationship between hardware components in a computer.

Key characteristics:

- Programs are executed sequentially through a fetch-decode-execute cycle.
- Uses a single memory space and a single path for both program instructions and data. The computer differentiates them based on their location, not their inherent type.
- Typically employs a single processor.

Most modern computers are based on these principles, though variations exist.

Von Neumann bottleneck

The fundamental limitation of the Von Neumann architecture is that instructions are processed sequentially, one at a time.

While improvements like more RAM, cache, or faster components can be made, significant gains in CPU performance are constrained by this sequential nature. This inherent limitation, where the sequential processing of instructions holds back overall efficiency, is known as the 'Von Neumann bottleneck'.